

Limbajul IMP

IMP este un limbaj IMPerativ foarte simplu.

Ce conține

- Expresii
 - Aritmetice
 - Booleene
 - Blocuri de instrucțiuni
 - De atribuire
 - Condiționale
 - De ciclare
- ```
x := 10;
sum := 0;
while (0 <= x) do (
 sum := sum + x;
 x := x + -1
)
```

## Ce nu conține

- Expresii cu efecte laterale
- Proceduri și funcții
- Schimbări abrupte de control

# Sintaxă formală

Sintaxa BNF a limbajului IMP

$E ::= \mathbb{Z} \mid L$   
 $\mid E + E \mid E - E \mid E * E$

$B ::= true \mid false$   
 $\mid E \leq E \mid E = E$   
 $\mid \neg B \mid B \wedge B \mid B \vee B$

$C ::= skip$   
 $\mid C ; C$   
 $\mid L := E$   
 $\mid \text{if } B \text{ then } C \text{ else } C$   
 $\mid \text{while } B \text{ do } C$   
 $\mid ( C )$

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# Semantica Evaluării a lui IMP

Expresii aritmetice

(NUM)  $\langle i, \sigma \rangle \Downarrow \langle i \rangle$

(ID)  $\langle x, \sigma \rangle \Downarrow \langle i \rangle$  dacă  $i = \sigma(x)$

(ADD)  $\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 + a_2, \sigma \rangle \Downarrow \langle i \rangle}$  dacă  $i = i_1 +_{Int} i_2$

(SUB)  $\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 - a_2, \sigma \rangle \Downarrow \langle i \rangle}$  dacă  $i = i_1 -_{Int} i_2$

(MUL)  $\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 * a_2, \sigma \rangle \Downarrow \langle i \rangle}$  dacă  $i = i_1 *_{Int} i_2$

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# Semantica Evaluării a lui IMP

Operatori de comparare

(LEQ-TRUE)  $\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 \leq a_2, \sigma \rangle \Downarrow \langle true \rangle}$  dacă  $i_1 \leq_{Int} i_2$

(LEQ-FALSE)  $\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 \leq a_2, \sigma \rangle \Downarrow \langle false \rangle}$  dacă  $i_1 >_{Int} i_2$

(EQ-TRUE)  $\frac{\langle a_1, \sigma \rangle \Downarrow \langle i \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i \rangle}{\langle a_1 = a_2, \sigma \rangle \Downarrow \langle true \rangle}$

(EQ-FALSE)  $\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 = a_2, \sigma \rangle \Downarrow \langle false \rangle}$  dacă  $i_1 \neq_{Int} i_2$

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## Semantica Evaluării a lui IMP

### Operatori logici

$$(\text{Bool}) \quad \langle t, \sigma \rangle \Downarrow \langle t \rangle$$

$$(\text{Not-True}) \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{false} \rangle}{\langle \neg b, \sigma \rangle \Downarrow \langle \text{true} \rangle}$$

$$(\text{And-True}) \quad \frac{\langle b_1, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle b_2, \sigma \rangle \Downarrow \langle t \rangle}{\langle b_1 \wedge b_2, \sigma \rangle \Downarrow \langle t \rangle}$$

$$(\text{And-False}) \quad \frac{\langle b_1, \sigma \rangle \Downarrow \langle \text{false} \rangle}{\langle b_1 \wedge b_2, \sigma \rangle \Downarrow \langle \text{false} \rangle}$$

$$(\text{Or-True}) \quad \frac{\langle b_1, \sigma \rangle \Downarrow \langle \text{true} \rangle}{\langle b_1 \vee b_2, \sigma \rangle \Downarrow \langle \text{true} \rangle}$$

$$(\text{Or-False}) \quad \frac{\langle b_1, \sigma \rangle \Downarrow \langle \text{false} \rangle \quad \langle b_2, \sigma \rangle \Downarrow \langle t \rangle}{\langle b_1 \vee b_2, \sigma \rangle \Downarrow \langle t \rangle}$$

$$(\text{Not-False}) \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle}{\langle \neg b, \sigma \rangle \Downarrow \langle \text{false} \rangle}$$

## Semantica Evaluării a lui IMP

### Instrucțiuni simple

$$(\text{Skip}) \quad \langle \text{skip}, \sigma \rangle \Downarrow \langle \sigma \rangle$$

$$(\text{ASGN}) \quad \frac{\langle a, \sigma \rangle \Downarrow \langle i \rangle}{\langle x := a, \sigma \rangle \Downarrow \langle \sigma' \rangle} \quad \text{dacă } \sigma' = \sigma_{x \mapsto i}$$

$$(\text{Seq}) \quad \frac{\langle s_1, \sigma \rangle \Downarrow \langle \sigma' \rangle \quad \langle s_2, \sigma' \rangle \Downarrow \langle \sigma'' \rangle}{\langle s_1 ; s_2, \sigma \rangle \Downarrow \langle \sigma'' \rangle}$$

$$(\text{Block}) \quad \frac{\langle s, \sigma \rangle \Downarrow \langle \sigma' \rangle}{\langle (s), \sigma \rangle \Downarrow \langle \sigma' \rangle}$$

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## Semantica Evaluării a lui IMP

### Blocuri și instrucțiuni de ciclare

$$(\text{If-True}) \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle bl_1, \sigma \rangle \Downarrow \langle \sigma_1 \rangle}{\langle \text{if } b \text{ then } bl_1 \text{ else } bl_2, \sigma \rangle \Downarrow \langle \sigma_1 \rangle}$$

$$(\text{If-False}) \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{false} \rangle \quad \langle bl_2, \sigma \rangle \Downarrow \langle \sigma_2 \rangle}{\langle \text{if } b \text{ then } bl_1 \text{ else } bl_2, \sigma \rangle \Downarrow \langle \sigma_2 \rangle}$$

$$(\text{While-True}) \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle bl, \sigma \rangle \Downarrow \langle \sigma' \rangle \quad \langle \text{while } b \text{ do } bl, \sigma' \rangle \Downarrow \langle \sigma'' \rangle}{\langle \text{while } b \text{ do } bl, \sigma \rangle \Downarrow \langle \sigma'' \rangle}$$

$$(\text{While-False}) \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{false} \rangle}{\langle \text{while } b \text{ do } bl, \sigma \rangle \Downarrow \langle \sigma \rangle}$$

## Semantica Evaluării a lui IMP

### Arbori de derivare

$$(\text{Seq}) \quad \frac{(\text{ASGN}) \quad \frac{(\text{Num}) \quad \langle 3, \mathbf{0} \rangle \Downarrow \langle 3 \rangle}{\langle a := 3, \mathbf{0} \rangle \Downarrow \langle a \mapsto 3 \rangle} \quad (\text{ASGN}) \quad \frac{\dots}{\langle a := a + 4, a \mapsto 3 \rangle \Downarrow \langle a \mapsto 7 \rangle}}{\langle a := 3; a := a + 4, \mathbf{0} \rangle \Downarrow \langle a \mapsto 7 \rangle}$$

$$(\text{ASGN}) \quad \frac{(\text{Add}) \quad \frac{(\text{Id}) \quad \langle a, a \mapsto 3 \rangle \Downarrow \langle 3 \rangle \quad (\text{Num}) \quad \langle 4, a \mapsto 3 \rangle \Downarrow \langle 4 \rangle}{\langle a + 4, a \mapsto 3 \rangle \Downarrow \langle 7 \rangle}}{\langle a := a + 4, a \mapsto 3 \rangle \Downarrow \langle a \mapsto 7 \rangle}$$

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# Semantica SOS a lui IMP

## Expresii aritmetice

- Un întreg este valoare — nu poate fi redex, deci nu avem regulă

$$(Ib) \langle x, \sigma \rangle \rightarrow \langle i, \sigma \rangle \text{ dacă } i = \sigma(x)$$

- Ordine nespecificată de evaluare a argumentelor la adunare:

$$\frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1 + a_2, \sigma \rangle \rightarrow \langle a'_1 + a_2, \sigma \rangle}$$

$$\frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle a_1 + a_2, \sigma \rangle \rightarrow \langle a_1 + a'_2, \sigma \rangle}$$

$$(ADD) \langle i_1 + i_2, \sigma \rangle \rightarrow \langle i, \sigma \rangle \text{ dacă } i = i_1 + i_2$$

- Regulile pentru înmulțire (Mul) sunt la fel

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# Semantica SOS a lui IMP

## Expresii Booleene. Constante și operatorii de comparație.

- Constantele Booleene sunt valori — nu pot fi redex

$$\frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1 \leq a_2, \sigma \rangle \rightarrow \langle a'_1 \leq a_2, \sigma \rangle}$$

$$\frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle a_1 \leq a_2, \sigma \rangle \rightarrow \langle a_1 \leq a'_2, \sigma \rangle}$$

$$(LEQ-FALSE) \langle i_1 \leq i_2, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle \text{ dacă } i_1 > i_2$$

$$(LEQ-TRUE) \langle i_1 \leq i_2, \sigma \rangle \rightarrow \langle \text{true}, \sigma \rangle \text{ dacă } i_1 \leq i_2$$

- Regulile pentru egalitate (Eq) sunt la fel

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# Semantica SOS a lui IMP

## Expresii Booleene. Negația logică

$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \neg b, \sigma \rangle \rightarrow \langle \neg b', \sigma \rangle}$$

$$(NEG-TRUE) \langle \neg \text{true}, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

$$(NEG-FALSE) \langle \neg \text{false}, \sigma \rangle \rightarrow \langle \text{true}, \sigma \rangle$$

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# Semantica SOS a lui IMP

## Expresii Booleene. Și-ul logic

- Observați că evaluarea e scurtcircuitată

$$\frac{\langle b_1, \sigma \rangle \rightarrow \langle b'_1, \sigma \rangle}{\langle b_1 \wedge b_2, \sigma \rangle \rightarrow \langle b'_1 \wedge b_2, \sigma \rangle}$$

$$(AND-FALSE) \langle \text{false} \wedge b_2, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

$$(AND-TRUE) \langle \text{true} \wedge b_2, \sigma \rangle \rightarrow \langle b_2, \sigma \rangle$$

- Regulile pentru disjuncție (Or) sunt asemănătoare

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- Instrucțiunea skip este „valoarea” blocurilor și instrucțiunilor
- Blocurile evaluează pas cu pas instrucțiunile ce le conțin

$$\frac{\langle s, \sigma \rangle \rightarrow \langle s', \sigma' \rangle}{\langle (s), \sigma \rangle \rightarrow \langle (s'), \sigma' \rangle}$$

$$(BLOCK) \langle (skip), \sigma \rangle \rightarrow \langle skip, \sigma \rangle$$

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$$\frac{\langle s_1, \sigma \rangle \rightarrow \langle s'_1, \sigma' \rangle}{\langle s_1 ; s_2, \sigma \rangle \rightarrow \langle s'_1 ; s_2, \sigma' \rangle}$$

$$(SEQ) \langle skip ; s_2, \sigma \rangle \rightarrow \langle s_2, \sigma \rangle$$

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$$\frac{\langle a, \sigma \rangle \rightarrow \langle a', \sigma \rangle}{\langle x := a, \sigma \rangle \rightarrow \langle x := a', \sigma \rangle}$$

$$(ASGN) \langle x := i, \sigma \rangle \rightarrow \langle skip, \sigma' \rangle \text{ dacă } \sigma' = \sigma_{x \mapsto i}$$

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$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } b \text{ then } bl_1 \text{ else } bl_2, \sigma \rangle \rightarrow \langle \text{if } b' \text{ then } bl_1 \text{ else } bl_2, \sigma \rangle}$$

$$(IF-TRUE) \langle \text{if true then } bl_1 \text{ else } bl_2, \sigma \rangle \rightarrow \langle bl_1, \sigma \rangle$$

$$(IF-FALSE) \langle \text{if false then } bl_1 \text{ else } bl_2, \sigma \rangle \rightarrow \langle bl_2, \sigma \rangle$$

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# Semantica SOS a lui IMP

Instrucțiunea de ciclare

$$\begin{aligned} \text{(WHILE)} \quad \langle \text{while } b \text{ do } bl, \sigma \rangle &\rightarrow \\ &\langle \text{if } b \text{ then } (bl ; \text{while } b \text{ do } bl) \text{ else skip}, \sigma \rangle \end{aligned}$$

# Semantica SOS a lui IMP

Execuție pas cu pas

$$\begin{aligned} &\langle \text{while } 0 \leq i \text{ do } i := i + -4, i \mapsto 3 \rangle && \xrightarrow{\text{WHILE}} \\ &\langle \text{if } 0 \leq i \text{ then } (i := i + -4 ; \text{while } 0 \leq i \text{ do } i := i + -4) \text{ else skip}, i \mapsto 3 \rangle && \xrightarrow{\text{ID}} \\ &\langle \text{if } 0 \leq 3 \text{ then } (i := i + -4 ; \text{while } 0 \leq i \text{ do } i := i + -4) \text{ else skip}, i \mapsto 3 \rangle && \xrightarrow{\text{LEQ-TRUE}} \\ &\langle \text{if } (\text{true}) (i := i + -4 ; \text{while } 0 \leq i \text{ do } i := i + -4) \text{ else skip}, i \mapsto 3 \rangle && \xrightarrow{\text{IF-TRUE}} \\ &\langle (i := i + -4 ; \text{while } 0 \leq i \text{ do } i := i + -4), i \mapsto 3 \rangle && \xrightarrow{\text{ID}} \end{aligned}$$

# Semantica SOS a lui IMP

Execuție pas cu pas

$$\begin{aligned} &\langle (i := 3 + -4 ; \text{while } 0 \leq i \text{ do } i := i + -4), i \mapsto 3 \rangle && \xrightarrow{\text{ADD}} \\ &\langle (i := -1 ; \text{while } 0 \leq i \text{ do } i := i + -4), i \mapsto 3 \rangle && \xrightarrow{\text{ASGN}} \\ &\langle (\text{skip} ; \text{while } 0 \leq i \text{ do } i := i + -4), i \mapsto -1 \rangle && \xrightarrow{\text{SEQ}} \\ &\langle (\text{while } 0 \leq i \text{ do } i := i + -4), i \mapsto -1 \rangle && \xrightarrow{\text{WHILE}} \\ &\langle (\text{if } 0 \leq i \text{ then } (i := i + -4 ; \text{while } 0 \leq i \text{ do } i := i + -4) \text{ else skip}), i \mapsto -1 \rangle && \xrightarrow{\text{ID}} \\ &\langle (\text{if } 0 \leq -1 \text{ then } (i := i + -4 ; \text{while } 0 \leq i \text{ do } i := i + -4) \text{ else skip}), i \mapsto -1 \rangle && \xrightarrow{\text{LEQ-FALSE}} \end{aligned}$$

# Semantica SOS a lui IMP

Execuție pas cu pas

$$\begin{aligned} &\langle (\text{if false then } (i := i + -4 ; \text{while } 0 \leq i \text{ do } i := i + -4) \text{ else skip}), i \mapsto -1 \rangle && \xrightarrow{\text{IF-FALSE}} \\ &\langle (\text{skip}), i \mapsto -1 \rangle && \xrightarrow{\text{BLOCK}} \\ &\langle \text{skip}, i \mapsto -1 \rangle \end{aligned}$$